

AP Precalculus: 2026–2027

Unit 3: Trigonometric and Polar Functions Lesson 3.10: Trigonometric Equations and Inequalities (3–4 instructional periods, 55 minutes each)

Primary Objective: Students will solve equations and inequalities involving trigonometric functions by isolating the trig function, applying inverse trigonometric functions to find the principal value, and using periodicity to generate the general solution. In contextual scenarios, students will identify the implied domain restriction and select only the solutions that are valid in context. (*Big Idea: Functions*)

Priority Ideas & Skills

AP Skills

1.A Solve equations/inequalities analytically — Isolate a trigonometric function, apply an inverse trig function to find the principal value, then use periodicity to write the general solution.

2.A Identify information from representations — Read solution intervals from a contextual function model (graph or formula) and connect them to the question being asked.

3.B Apply numerical results — Interpret solutions in context: select values of t (or θ) that lie within the implied domain of the scenario.

Key Understandings (EKs)

EK 3.10.A.1: Inverse trigonometric functions are useful in solving equations and inequalities involving trigonometric functions, but solutions may need to be modified due to domain restrictions of the inverse function.

EK 3.10.A.2: Because trigonometric functions are periodic, there are often infinitely many solutions to trigonometric equations when the domain is unrestricted.

EK 3.10.A.3: In trigonometric equations and inequalities arising from a contextual scenario, there is often an implied domain restriction that limits the number of valid solutions.

Vocabulary, Concepts, & Theorems

Term

Definition / Note

trigonometric equation	An equation in which the unknown appears inside a trigonometric function (sin, cos, tan, etc.).
principal solution	The solution obtained by applying an inverse trig function directly; restricted to the range of that inverse function.
general solution	All solutions to a trigonometric equation, expressed using the period (e.g. $\theta = \frac{\pi}{6} + 2\pi k$, $k \in \mathbb{Z}$).
domain restriction	A limit on the variable imposed by the context of the problem (e.g. $0 \leq t \leq 24$ for a 24-hour tide model).

Activate Prior Knowledge & Spiral Review

Spiral topics activated during Warm-Up (first 8 minutes):

- Inverse trig values:** Evaluate $\arcsin(\frac{\sqrt{2}}{2})$ and $\arccos(-\frac{1}{2})$ — confirms students recall the ranges of arcsin and arccos before the lesson relies on them.
- Unit circle / reference angles:** Given $\sin \theta = \frac{1}{2}$, find all $\theta \in [0, 2\pi)$ — prerequisite for generating the second solution family beyond the principal value.
- Isolating a trig function:** Solve $2 \sin x + 1 = 0$ by isolating $\sin x$ (no inverse trig yet) — reinforces the algebraic manipulation that opens every trig equation.

Hook — Entry Event

Scenario: Display the graph of $h(t) = 3 \sin(\frac{\pi}{6}t) + 5$, representing tidal height (in feet) over a 24-hour period.

Discussion prompts:

1. “A harbor requires at least 7 feet of water for a ship to dock. During what hours today can the ship safely enter?”
2. “If the tide is exactly 7 feet, how many times does that happen in one 24-hour period? Could it happen again tomorrow?”
3. “How is this question different from ‘find the maximum tide’? What new mathematical tool do we need?”

Key tension to surface: Reading a graph gives approximate answers. Finding exact times requires solving $3\sin(\frac{\pi}{6}t) + 5 = 7$ analytically. Today’s tools — inverse trig functions plus periodicity — give exact solutions and also reveal the full interval where the inequality $h(t) \geq 7$ holds.

Transition: “By the end of this lesson you will solve that equation exactly, write all solutions as a general formula, and determine precisely which hours the ship can enter.”

Lesson — Instruction Sequence

Step 1 — Infinitely Many Solutions (10 min)

Motivate EK 3.10.A.2 with $\sin \theta = \frac{1}{2}$. On the unit circle, mark both solutions in $[0, 2\pi)$: $\theta = \frac{\pi}{6}$ and $\theta = \frac{5\pi}{6}$. Ask: “Can we add 2π and get another solution?” Yes — leading to infinitely many in an unrestricted domain. Write the *general solution* in two families: $\theta = \frac{\pi}{6} + 2\pi k$ and $\theta = \frac{5\pi}{6} + 2\pi k$, $k \in \mathbb{Z}$. Emphasize: the $2\pi k$ comes from the period of sine; each trig function contributes its own periodicity to the general solution.

Step 3 — Contextual Domain Restriction (12 min)

Return to the hook model $h(t) = 3\sin(\frac{\pi}{6}t) + 5$. Solve the equation $h(t) = 7$:

1. Isolate: $\sin(\frac{\pi}{6}t) = \frac{2}{3}$.
2. Let $u = \frac{\pi}{6}t$; solve $\sin u = \frac{2}{3}$.
3. Principal: $u = \arcsin(\frac{2}{3})$; second: $u = \pi - \arcsin(\frac{2}{3})$.
4. Solve for t ; apply domain $t \in [0, 24]$ to select valid values.

Then solve $h(t) \geq 7$ by identifying the interval between the two boundary solutions where sine exceeds $\frac{2}{3}$. Emphasize EK 3.10.A.3: context ($0 \leq t \leq 24$) means we keep only the solutions in that window.

Pacing note: Steps 1–2 in Periods 1–2; Steps 3–4 + activity in Periods 3–4.

Step 2 — Inverse Trig Procedure (12 min)

Teach the four-step algorithm (EK 3.10.A.1) using $2\cos \theta - 1 = 0$:

1. Isolate: $\cos \theta = \frac{1}{2}$.
2. Principal value: $\theta = \arccos(\frac{1}{2}) = \frac{\pi}{3}$.
3. Second solution in $[0, 2\pi)$: $\theta = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$ (cosine is positive in Q1 and Q4, so reflect over the x -axis).
4. General solution: $\theta = \frac{\pi}{3} + 2\pi k$ or $\theta = \frac{5\pi}{3} + 2\pi k$, $k \in \mathbb{Z}$.

Discuss: the second solution in Step 3 depends on which quadrants give the correct sign of the trig function — this is why knowing the unit circle is essential.

Step 4 — Trig Inequalities (10 min)

Generalize the structure for $\sin \theta \geq c$ ($|c| \leq 1$):

- Find the two boundary solutions in one period.
 - Sine $\geq c$ on the arc between those values (the “upper” arc of the unit circle).
 - Extend using $2\pi k$ if the domain is unrestricted.
- Briefly contrast $\cos \theta \leq c$ (solution on the “lower” arc) and $\tan \theta > c$ (solution spanning half a period). Close: “Why does the contextual domain sometimes give you fewer solutions than the general formula?”

Active Monitoring

- **Missing second solution:** Students often find only the principal value and stop. Prompt: “Sine (or cosine) takes this value in two quadrants per period — which other angle gives the same value?”
- **Forgetting $+2\pi k$:** After finding two solutions in one period, students omit the general form. Ask: “How many solutions exist if the domain is all real numbers?”
- **Domain restriction errors:** Students list general-solution values without filtering. Prompt: “Does this t

value fall within the 24-hour context? Is it physically meaningful?"

- **Inequality direction:** Reinforce that isolating the trig function (dividing by a positive coefficient) does not flip the inequality sign; direction flips only when dividing by a negative.

Group Work & Differentiation

Students work in groups of 3–4 on the tiered activity (assign by teacher or allow self-selection with guidance).

- **Tier R (Remediate):** General solution formulas are provided. Students substitute specific integers $k = 0, 1, 2, -1, -2$ to list all solutions in $[0, 4\pi]$, then circle those in a specified sub-interval. Step-by-step scaffold included.
- **Tier A (Apply):** Students solve three trig equations without scaffolding: one straightforward, one requiring a quadratic approach ($u = \sin \theta$), and one contextual problem requiring a domain restriction. Students find all solutions in the given interval.
- **Tier E (Extend):** Students solve a trig inequality using a table of pre-computed values of $f(\theta) = 2 \cos \theta - 1$ to identify where $f(\theta) > 0$, then write the solution as a union of intervals. One contextual problem requires selecting and interpreting valid solutions in writing.

Individual Work & Assessment

Exit ticket administered independently (final 5 minutes of Period 4).

1. Solve $\sin \theta = -\frac{\sqrt{3}}{2}$ for all $\theta \in [0, 2\pi]$.
2. Explain in one or two sentences why a trigonometric equation has infinitely many solutions when the domain is all real numbers.

Data use: Sort exit tickets into three groups — correct general solution (both families), correct principal value only (one family), incorrect. Use groupings to decide whether Lesson 3.11 opens with a brief recap of the general-solution structure.

Reinforcement & Extension

Homework (Lesson 3.10): 5–6 problems: solving basic trig equations for all solutions in a given interval, writing the general solution, one contextual domain problem with interpretation, 2 AP-style MC items, and an extension asking students to solve $2 \sin^2 \theta - \sin \theta - 1 = 0$ via quadratic substitution $u = \sin \theta$.

Preview of Lesson 3.11 — Secant, Cosecant, and Cotangent: Ask students to consider: “We have three more trig functions — sec, csc, and cot — defined as the reciprocals of cosine, sine, and tangent. How would you expect to solve an equation like $\sec \theta = 2$? What period would you predict?” Lesson 3.11 extends today’s skills to these reciprocal functions.